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Batch 15

C program

```
#include <stdio.h>

int pivot_index(int *arr, int size)
{
    int i;
    int totalSum = 0;
    int leftSum = 0;
    int rightSum;

    /* Calculate total sum of array */
    for (i = 0; i < size; i++)
    {
        totalSum = totalSum + *(arr + i);
    }

    /* Find the leftmost pivot index */
    for (i = 0; i < size; i++)
    {
        rightSum = totalSum - leftSum - *(arr + i);

        if (leftSum == rightSum)
        {
            return i;
        }

        leftSum = leftSum + *(arr + i);
    }

    return -1;
}

int main()
{
```

```
    int size, i, index;

    printf("Enter the size of the array: ");
    scanf("%d", &size);

    if (size <= 0)
    {
        printf("Invalid array size.\n");
        return 0;
    }

    /* Variable-length array */
    int arr[size];

    printf("Enter %d array elements:\n", size);

    for (i = 0; i < size; i++)
    {
        scanf("%d", &arr[i]);
    }

    /* Pass array address to the function */
    index = pivot_index(arr, size);

    if (index == -1)
    {
        printf("-1\n");
    }
    else
    {
        printf("Pivot index: %d\n", index);
    }

    return 0;
}
```

Algorithm

1. Start.
2. Read the size `size` of the array.
3. If `size <= 0`, display an invalid-size message and terminate.
4. Declare a variable-length array `arr[size]`.
5. Read all the array elements.
6. Pass the address of `arr[]` and `size` to `pivot_index()`.
7. Inside `pivot_index()`:
    - Calculate the sum of all elements and store it in `totalSum`.

- Initialize `leftSum = 0`.
- Starting from index `0`, calculate:
  `rightSum = totalSum - leftSum - arr[i]`
- Compare `leftSum` and `rightSum`.
- If they are equal, return the current index immediately. This ensures the **leftmost** pivot index is returned.
- Otherwise, add the current element to `leftSum`.

8. If no index satisfies the condition, return `-1`.
9. In `main()`, display the pivot index if found; otherwise display `-1`.
10. Stop.

Test Case Table

Test	Input Array	Expected Output	Reason
1	`1 7 3 6 5 6`	`Pivot index: 3`	`1+7+3 = 5+6 = 11`
2	`1 2 3`	`-1`	No equal left and right sums
3	`2 1 -1`	`Pivot index: 0`	Left = `0`, Right = `1 + (-1) = 0`
4	`1 2 1`	`Pivot index: 1`	Left = `1`, Right = `1`
5	`0`	`Pivot index: 0`	Left = `0`, Right = `0`
6	`1 -1 0`	`Pivot index: 2`	Left = `1 + (-1) = 0`, Right = `0`
7	`0 0 0`	`Pivot index: 0`	Leftmost valid pivot is index `0`

Execution

```
Last login: Thu Oct 15 21:52:17 on ttys000
bhavya@Bhavyas-MacBook-Air ~ % nano day48.c
bhavya@Bhavyas-MacBook-Air ~ % clang day48.c -o day48
bhavya@Bhavyas-MacBook-Air ~ % ./day48
Enter the size of the array: 6
Enter 6 array elements:
1 7 3 6 5 6
Pivot index: 3
bhavya@Bhavyas-MacBook-Air ~ % ./day48
Enter the size of the array: 3
Enter 3 array elements:
1 2 3
-1
bhavya@Bhavyas-MacBook-Air ~ % ./day48
Enter the size of the array: 3
Enter 3 array elements:
2 1 -1
Pivot index: 0
bhavya@Bhavyas-MacBook-Air ~ %
```